

2D 270° NAVIGATION LIDAR · LR-1BS SERIES

COST-DOWN OLEI LR-1BS5D-C1 270° LIDAR



■ WHY THIS SENSOR

- **Built on the proven 1BS platform** — same sensing architecture, specified down to industrial and commercial cost targets
- **0.2–25 m at 80% reflectivity** — 0.2–8 m at 10% and 0.2–2 m at 4%, plus 25 m on reflectors
- **< ±20 mm absolute accuracy on common targets** — < ±40 mm on reflectors, ≤ 20 mm repeat
- **Point cloud over 100 Mbps Ethernet** — azimuth, distance and signal strength, plus one switching output
- **M12 connectors on 25 cm pigtails** — 5-pin A-coded male for power, 4-pin D-coded female for LAN
- **85 mm body height, 60 × 60 mm footprint** — 92.4 mm overall including the connector plugs
- **Tested and certified for multiple reliability standards** — designed to industrial standards for vibration, shock and EMC
- **450 g, IP65, 5 W at 12–24 V DC** — OLAMViewer and a ROS driver, bottom or back mounting

0.2–25 m

MEASUREMENT RANGE

270°

FIELD OF VIEW

< ±20 mm

ABSOLUTE ACCURACY

40K

POINTS / SECOND

■ APPLICATION SCENARIOS



AMR & AGV

270° obstacle avoidance from a single chassis-mounted scanner



COMPACT PLATFORMS

Service robots and small AMRs with tight sensor pockets



OEM INTEGRATION

Volume platforms where size, weight and power are budgeted



INTRALOGISTICS

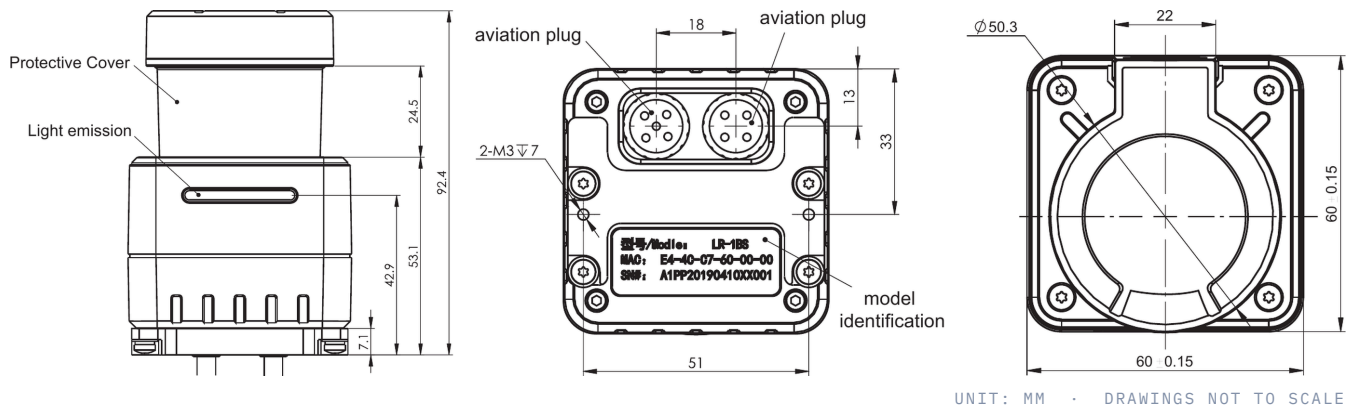
Forklifts, tuggers and conveyor zones in active aisles

LR-1BS5D-C1 — SPECIFICATIONS

PARAMETER	TECHNICAL DATA
Sensing technology	2D scanning ToF LiDAR
Laser wavelength	905 ± 20 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Laser channels	1
Field of view (horizontal)	270°
Beam divergence	5.8 mrad longitudinal 0.58 mrad transversal
Scanning frequency	10–25 Hz
Angle resolution (horizontal)	0.225°
Measurement range (based on reflectivity)	0.2–25 m @ 80% reflectivity 0.2–8 m @ 10% · 0.2–2 m @ 4% · 0.2–25 m @ reflector
Accuracy ①	Common targets: absolute < ±20 mm Reflector: absolute < ±40 mm · Repeat ≤ 20 mm
Distance resolution	1 mm
Signal strength	0–60000
Scanning rate	16K–40K pts/s
Horizontal error (horizontal angle)	≤ ±1°

PARAMETER	TECHNICAL DATA
Switching outputs	1
Ethernet connection	100 Mbps Ethernet
Connection	5-pin M12×1 A-coded male, 25 cm (power) 4-pin M12×1 D-coded female socket, 25 cm (LAN)
Operating voltage	12–24 V DC
Power consumption @ 25 °C	5 W
Mechanical dimensions	60 × 60 × 92.4 mm
Weight	450 g
Degree of protection	IP65 or optional
Operating temperature	–10 to +50 °C
Storage temperature	–30 to +70 °C
Air humidity	< 95% RH
Indicator status	RGB × 4
Ambient light limit	80 000 lux
Mounting	Bottom / back
Basic software	OLAMViewer · ROS driver

■ OUTLINE AND MOUNTING



Certification. RoHS, REACH, EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. ① Standard value; actual value depends on environmental conditions. The manufacturer's feature summary quotes up to 30 m effective detection on an 80% reflectivity white board, while the specification table gives 0.2–25 m @ 80% — quote the table. Typical range for objects up to 10% remission is 8 m. The power lead accepts DC 12–30 V per the cable definition, while the specification table states 12–24 V DC; design to 12–24 V.

Power cable definition. 1 Red — VCC (DC 12–30 V) · 2 Blue — NC · 3 Black — GND · 4 Brown — NC · 5 Grey — NC

Installation. Mount the scanner so the 270° plane is unobstructed through its full sweep and the blind 90° faces structure, not traffic. Both leads are 25 cm pigtailed terminating in M12 plugs, so plan a junction immediately behind the mount. Keep the protective cover clear of spray and dust and clean it on a schedule in dirty environments.

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